B1 Linear relations and equations: problems

← Module overview · Solutions

Problems 1–13 are practice, 14–22 are at the level of a VWO school test (*toets*), and 23–30 are challenge problems. Unless a problem says otherwise, give exact answers (fractions, not rounded decimals) and show your working.

Practice

1. Given the formula $y = 3x - 5$.

(a) Make a table for $x = -2, -1, 0, 1, 2, 3$.
(b) Give the slope (*richtingscoëfficiënt*) and the start value (*startgetal*).
(c) Does the point $(4, 7)$ lie on the graph? Show how you know.

2. (a) Calculate the slope of the line through $(2, 5)$ and $(6, 17)$.

(b) Give the formula of this line.

3. Find the formula of the line through $(-3, 4)$ and $(5, -12)$. Check your answer with both points.

4. Find the formula of the line with slope $\frac{1}{2}$ that passes through $(4, -1)$.

5. Line k has formula $y = -3x + 7$. Find the formula of the line parallel to k through the point $(2, 4)$.

6. Solve, using the balance method (*balansmethode*).

(a) $5x - 7 = 2x + 8$
(b) $3(2x - 1) - 4(x + 2) = 7$
(c) $\dfrac{x + 1}{3} - \dfrac{x - 2}{4} = 2$
(d) $0.4x + 1.2 = 0.1x - 0.6$

7. For each equation, decide whether it has no solution, exactly one solution, or every number as a solution. Explain.

(a) $2(x + 3) = 2x + 5$
(b) $3(x - 2) + x = 4x - 6$

8. Solve these inequalities (*ongelijkheden*).

(a) $3x - 4 < 11$
(b) $5 - 2x \geq 13$
(c) $\dfrac{x - 1}{-3} > 2$
(d) $2(x - 3) - 5x \leq x + 10$

9. Solve by substitution (*substitueren*):

$$\begin{cases} y = 2x - 3 \\ 3x + 2y = 15 \end{cases}$$

10. Solve by elimination (*combineren*):

$$\begin{cases} 3x + 4y = 10 \\ 5x - 2y = 34 \end{cases}$$

11. Draw the lines $y = x + 1$ and $y = -2x + 7$ in one coordinate system. Read off the intersection point (*snijpunt*) and check it by calculation.

12. For each system, decide (before solving) whether it has no solution, exactly one solution or infinitely many solutions. Then find all solutions.

(a) $2x - 3y = 6$ and $4x - 6y = 12$
(b) $2x - 3y = 6$ and $4x - 6y = 10$
(c) $2x - 3y = 6$ and $3x + 2y = 9$

13. Find the formula of the line perpendicular (*loodrecht*) to $y = \frac{2}{3}x + 1$ that passes through $(4, 1)$.

School-exam level

14. Phone plans. A provider offers three plans for calling.

- Plan A: €15 per month plus €0.10 per minute.
- Plan B: €5 per month plus €0.25 per minute.
- Plan C: €30 per month, unlimited calling.

(a) Give a formula for the monthly cost K (in euros) of plans A and B in terms of the number of minutes t.
(b) Calculate the costs of all three plans for 40 minutes and for 120 minutes.
(c) Using an inequality, calculate from how many whole minutes plan A is cheaper than plan B.
(d) For which whole numbers of minutes is plan A the cheapest of the three? Explain.

15. Taxi. A taxi company charges a fixed start fee plus a fixed price per kilometre. A ride of 8 km costs €22.10 and a ride of 13 km costs €32.60.

(a) Show that the price per kilometre is €2.10 and calculate the start fee.
(b) Mila has €50. How many whole kilometres can she travel at most?
(c) A competitor charges €3.50 plus €2.30 per km. For which distances is the competitor cheaper?

16. Temperature. In the Netherlands temperature is measured in degrees Celsius (C), in the United States in degrees Fahrenheit (F). Water freezes at $0\,^\circ\text{C} = 32\,^\circ\text{F}$ and boils at $100\,^\circ\text{C} = 212\,^\circ\text{F}$, and the relation is linear.

(a) Show that $F = 1.8C + 32$.
(b) Body temperature is $37\,^\circ$C. Convert to °F.
(c) At which temperature do the two scales show the same number?
(d) Scientists use kelvin: $K = C + 273.15$. Give a formula for F in terms of K, in the form $F = aK + b$.

17. Bake sale. For a charity event, a class bakes cupcakes. They pay €120 for the hire of an oven and a stand, plus €0.80 in ingredients per cupcake. They sell the cupcakes for €2.30 each. Assume they sell every cupcake they bake.

(a) Show that the profit W (in euros) for n cupcakes is $W = 1.5n - 120$.
(b) Calculate the break-even point.
(c) How many cupcakes must they sell to make a profit of at least €200?

18. Candles. Candle A is 30 cm long and burns 2 cm per hour. Candle B is 22 cm long and burns 1.2 cm per hour. Both are lit at the same moment.

(a) Give formulas for the lengths L_A and L_B (in cm) after t hours.
(b) After how many hours are the candles equally long? How long are they then?
(c) Which candle burns out first, and how much later does the other one burn out? Give your answer in hours and minutes.

19. Triangle. Given the points $A(-2, 1)$, $B(4, 4)$ and $C(1, -5)$.

(a) Find the formula of line AB.
(b) Find the formula of the line through C parallel to AB.

(c) Find the formula of the line through C perpendicular to AB.
(d) Calculate the intersection point of the line in (c) with line AB. What does your answer tell you about triangle ABC?

20. Cinema. An adult ticket costs €12.00 and a child ticket €7.50. On Saturday afternoon 85 tickets were sold, for a total of €858. Set up a system of equations and calculate how many adult tickets and how many child tickets were sold.

21. Saving. Sven has €40 and saves €6.50 every week. His sister Noor has €95 and spends €4 of it every week. From which week on does Sven have more money than Noor? Solve with an inequality and check your answer.

22. Water tanks. Tank A contains 500 litres and is drained at 12 litres per minute. At the same moment, tank B, which contains 80 litres, is filled at 9 litres per minute. Let t be the time in minutes.

(a) Give formulas for the volumes V_A and V_B.
(b) When do the tanks contain the same amount, and how much is that?
(c) During which time interval does tank A contain more than twice as much as tank B? (Take into account when tank A is empty.)

Challenge

23. Parameters. Consider the system

$$\begin{cases} 2x + py = 5 \\ 4x + 6y = q \end{cases}$$

(a) For which values of p and q does the system have no solution?
(b) For which values does it have infinitely many solutions?
(c) For which values does it have exactly one solution? Give that solution in terms of p and q.

24. The perpendicular rule, a second proof. Let ℓ_1 and ℓ_2 be the lines through the origin with slopes a_1 and a_2 (both not 0). Use the points $O(0,0)$, $P(1,a_1)$ and $Q(1,a_2)$ and Pythagoras' theorem (including its converse) to prove that $\ell_1 \perp \ell_2$ if and only if $a_1 a_2 = -1$. (You may use that the squared distance between (x_1,y_1) and (x_2,y_2) is $(x_2-x_1)^2+(y_2-y_1)^2$.)

25. Whole-number solutions.

(a) Find all pairs of positive whole numbers (x,y) with $3x + 5y = 47$. Explain why you have found them all.
(b) A shop sells stickers only in packs of 3 and packs of 5. Prove that you can buy exactly n stickers for every whole number $n \geq 8$, and find all $n \geq 1$ that are impossible.

26. Three unknowns. Solve

$$\begin{cases} x + y + z = 6 \\ 2x - y + z = 3 \\ x + 2y - z = 2 \end{cases}$$

27. Why elimination is safe. Let E_1 and E_2 be two linear equations in x and y.

(a) Prove that for every number k the systems $\{E_1, E_2\}$ and $\{E_1, E_2 + kE_1\}$ have exactly the same solutions.
(b) Prove that for every number $m \neq 0$, the systems $\{E_1, E_2\}$ and $\{mE_1, E_2\}$ have exactly the same solutions.
(c) Give a concrete example showing that (b) fails for $m = 0$.

28. A family of lines. For every number p, line ℓ_p has formula $y = px + 2p - 3$.

(a) Prove that all lines ℓ_p pass through one common point, and find it.

(b) For which values of p does ℓ_p intersect the segment of the x-axis from $(0,0)$ to $(4,0)$ (endpoints included)?

29. Digits. A two-digit number has digit sum 11. If you swap the two digits, the number increases by 27. Find the number, and prove it is the only one.

30. Three lines. The lines $y = 2x + 1$, $y = -x + 7$ and $y = \frac{1}{2}x + k$ pass through one common point.

(a) Find k.
(b) Calculate the area of the triangle formed by the lines $y = 2x + 1$, $y = -x + 7$ and the x-axis.